

Question	Answer	Marks
7(a)	number of protons = 92	A1
	number of neutrons = 142	A1
7(b)	$5.6 \text{ MeV} = 5.6 \times 1.60 \times 10^{-19} \times 10^6 \text{ (= } 8.96 \times 10^{-13} \text{ J)}$	C1
	number = $0.15 / (5.6 \times 1.60 \times 10^{-13})$ $= 1.7 \times 10^{11}$	A1
	or	
	$0.15 \text{ W} = 0.15 / (1.60 \times 10^{-19} \times 10^6) \text{ (= } 9.38 \times 10^{11} \text{ MeV s}^{-1})$	(C1)
	number = $9.38 \times 10^{11} / 5.6$ $= 1.7 \times 10^{11}$	(A1)